

CE 4011 — Final Project Report

A 2D Structural Analysis Program (Direct Stiffness Method) with a PyQt6 GUI

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Program: MSc Structural Engineering **Submission date:** [date] **Repository:** [GitHub URL]

1. Introduction

This project is a two-dimensional structural-analysis program built on the **Direct Stiffness Method (DSM)**. It analyses plane frames and trusses under mechanical loads, thermal loads, and prescribed support settlements, and it reports displacements, member internal forces, and support reactions together with a full set of visual diagrams.

The software has two faces over one shared engine:

- a **pure-Python computational engine** (NumPy/SciPy, no GUI dependency), and
- an interactive **PyQt6 desktop GUI** for drawing models and visualizing results, plus a **command-line interface** for reproducible batch runs.

The engine is written with object-oriented programming so that the two element types (frame and truss) share one interface but specialize their own mechanics. The design goal was correctness first (every result is checked against equilibrium and, where possible, closed-form solutions) and a clean separation between the analysis core and the presentation layer.

2. Scope of the software

Structural systems analysed

- **2D frames** (beam-columns with axial + flexural stiffness, 3 DOF/node: `ux`, `uy`, `rz`).
- **2D trusses** (axial-only members; rotational DOFs are omitted automatically at pure-truss nodes).
- **Mixed** frame/truss models in one structure.

Modelling features

- Point loads, full-length uniformly distributed loads (local / global / gravity direction), and nodal forces and moments.
- **Moment releases** (internal hinges) at member ends via static condensation.
- **Rigid end offsets** (rigid joint zones) on frame members.
- **Thermal loads**: frame members take top/bottom-fibre temperatures (mean → axial effect, gradient → bending); truss members take a uniform ΔT .
- **Support settlements** as prescribed restrained-DOF displacements.
- **Self-weight** as an optional automatic load.
- **Load cases and load combinations** (combinations formed by superposition).
- **Modal analysis** (natural frequencies / mode shapes) with consistent or lumped mass and user joint masses.

Outputs

- Nodal displacements, member end forces, support reactions.
- Bending-moment, shear-force, and axial-force diagrams; deformed shape.
- Equilibrium check and assembly/solve diagnostics.
- CSV export of internal forces at 21 stations per member.

Out of scope (honest boundaries): 3D analysis, material/geometric nonlinearity, dynamic time-history, P- Δ effects, and design-code checking.

3. Theoretical background

The program implements the **displacement (stiffness) method** for linear elastic skeletal structures.

Element stiffness. Each 2-node member has 6 local DOFs $[u_i, v_i, \theta_i, u_j, v_j, \theta_j]$. The frame element uses the standard beam-column matrix with EA/L axial terms and the $12EI/L^3$, $6EI/L^2$, $4EI/L$, $2EI/L$ flexural terms; the truss element keeps only the axial EA/L terms.

Transformation. A 6×6 rotation matrix $R = \text{diag}(T, T)$ (with the planar $T = \begin{bmatrix} c & s & 0 \\ -s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $c = \cos\theta$, $s = \sin\theta$) maps local to global: $k_{\text{global}} = R^T k_{\text{local}} R$, and likewise for the consistent load vector.

Assembly. Element contributions are scattered into the global stiffness matrix K and load vector F using each element's DOF address vector. Loads are stored as **energy-consistent equivalent nodal forces** p (added to F), and recovered with $q = k \cdot d - p$.

Boundary conditions and settlements. The system is partitioned into free (f) and restrained (s) DOFs:

$$\begin{bmatrix} K_{ff} & K_{fs} \\ K_{sf} & K_{ss} \end{bmatrix} \begin{bmatrix} D_f \\ D_s \end{bmatrix} = \begin{bmatrix} F_f \\ F_s + R \end{bmatrix}$$

Solving the free partition gives $K_{ff} \cdot D_f = F_f - K_{fs} \cdot D_s$, where D_s holds the prescribed support settlements (zero for ordinary supports). Reactions follow from $R = K \cdot D - F$ at the restrained DOFs. This treats settlement **physically** as a prescribed displacement, not as a fictitious load or a large spring.

Thermal loads. A restrained temperature change produces fixed-end forces. For a uniform change, $N_T = E \cdot A \cdot \alpha \cdot \Delta T$; for a through-depth gradient on a frame, $M_T = E \cdot I \cdot \alpha \cdot (t_{\text{bottom}} - t_{\text{top}}) / \text{depth}$. These enter as additional consistent load terms.

Singularity detection. Before solving, the free stiffness block is checked by SVD; a rank deficiency is reported as an instability with the dominant mechanism DOFs, rather than crashing.

Modal analysis. The generalized eigenproblem $K \varphi = \omega^2 M \varphi$ is solved (with massless rotational DOFs condensed out) for natural frequencies and mode shapes.

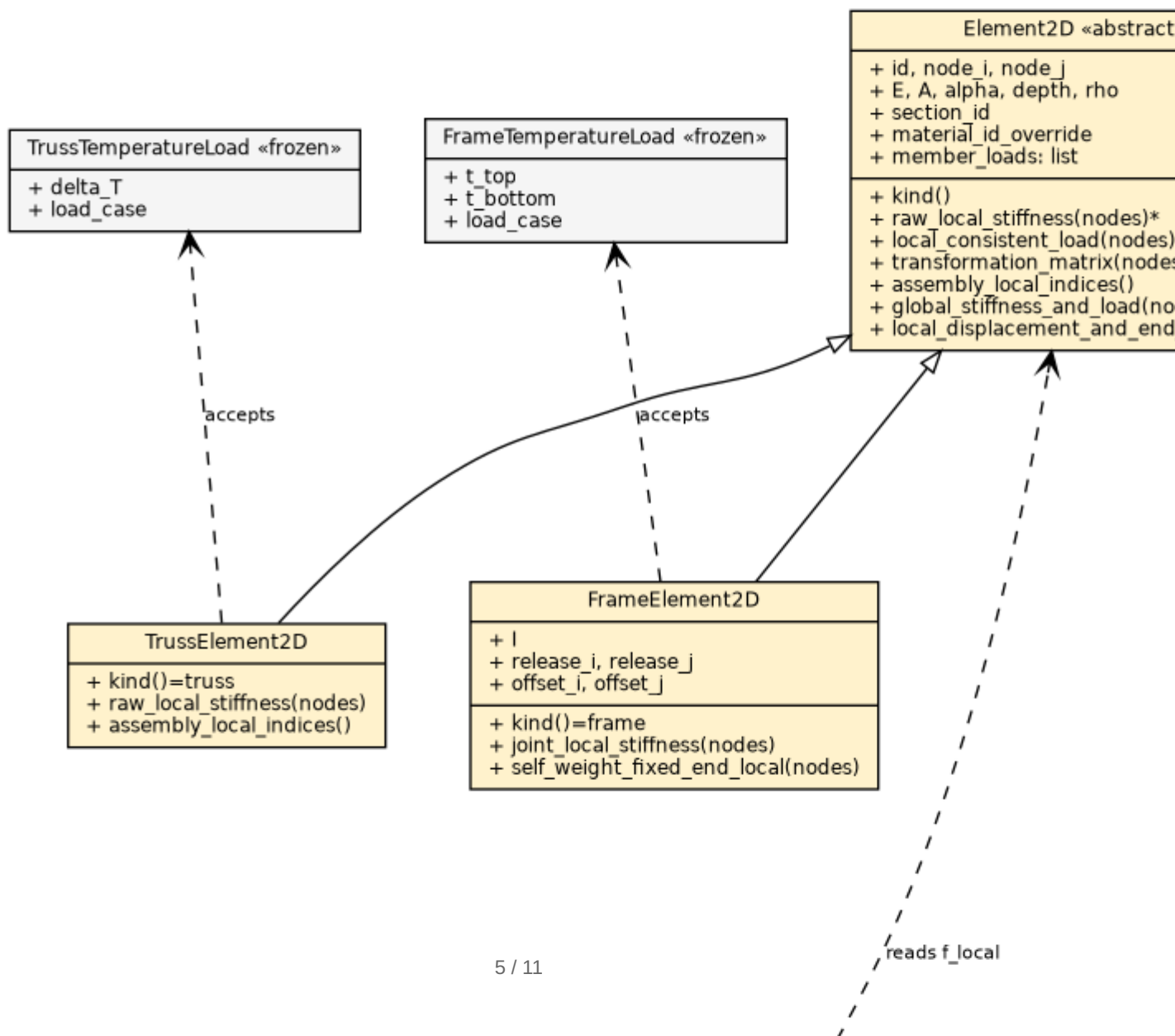
4. Software architecture and class diagrams

The program is organized into three strictly layered packages with a one-way dependency $\text{gui_qt} \rightarrow \text{gui_common} \rightarrow \text{engine}$:

Layer	Package	Responsibility
Computational engine	<code>structural_analysis/*.py</code>	model, elements, assembly, solve, post-processing, modal
Editor (Qt-free shared logic)	<code>structural_analysis/</code> <code>gui_common/</code>	undoable commands, pre-solve validation, file writer
GUI (presentation)	<code>structural_analysis/</code> <code>gui_qt/</code>	PyQt6 window, canvas, tools, dialogs, diagram drawing

(The assignment brief's `solver/` / `editor/` / `gui/` correspond to these three real packages; the mapping is documented in [docs/uml/architecture.md](#).)

The class diagram below is reproduced from [docs/uml/class_diagram.mmd](#) (a Graphviz version and rendered PNG are in [docs/uml/](#)).



4.1 Use of OOP in the computational engine

- **Element2D (abstract base)** defines the shared 6-DOF interface: `raw_local_stiffness`, `local_consistent_load`, `transformation_matrix`, `assembly_local_indices`, `global_stiffness_and_load`, and `local_displacement_and_end_forces`.
- **FrameElement2D** and **TrussElement2D** override these polymorphically: the frame builds the full flexural matrix, handles moment releases and rigid offsets, and accepts `FrameTemperatureLoad`; the truss builds an axial-only matrix, marks its rotational DOFs inactive, and accepts `TrussTemperatureLoad`.
- The assembler iterates over the abstract type and calls `element.global_stiffness_and_load(...)` without knowing the concrete subtype — this is the **polymorphism payoff**: one assembly loop serves every member kind, and a new element type would slot in by subclassing `Element2D`.
- **Type safety by design**: applying a `TrussTemperatureLoad` to a frame (or vice-versa) raises `TypeError`, so a modelling mistake fails loudly instead of producing a quietly wrong answer.

4.2 Major classes — purpose and relationships

Class / module	Responsibility	Key relationships
<code>StructuralModel</code>	aggregate root holding all model data	<i>composes</i> nodes, materials, sections, supports, elements, loads, cases
<code>Node</code> , <code>Material</code> , <code>Section</code> , <code>Support</code>	frozen value objects	<code>Section</code> → <code>Material</code> , <code>Element2D</code> → <code>Section</code>
<code>Element2D</code> → <code>FrameElement2D</code> , <code>TrussElement2D</code>	element mechanics	inherit <code>Element2D</code> ; read <code>Material/Section</code> data
load classes (<code>NodalLoad</code> , <code>UniformDistributedLoad</code> , <code>PointLoad</code> , <code>*TemperatureLoad</code>)	applied actions	attached to model / elements
<code>DofManager</code>	builds equation numbering, omits unused rotational DOFs	built from <code>StructuralModel</code> ; used by assembler/solver/post
<code>assembler</code>	validates connectivity, assembles <code>K</code> , <code>F</code>	uses <code>DofManager</code> , <code>Element2D</code>
<code>solver</code>	partitioned solve + SVD singularity check + settlements	uses <code>DofManager</code>
<code>postprocessor</code>	member forces, reactions, equilibrium check	uses <code>DofManager</code> , <code>StructuralModel</code>
<code>main</code>	orchestrates the pipeline; per-case filtering	calls <code>file_io</code> → <code>assembler</code> → <code>solver</code> → <code>postprocessor</code>
<code>AnalysisResult</code> / <code>MultiCaseAnalysisResult</code>	structured result containers; superposition	returned by <code>main</code>
<code>modal</code>	eigen-analysis	reads <code>StructuralModel</code> , mass matrices
<code>Command</code> (+ ~40 subclasses)	one undoable model mutation each	<code>do(model)</code> / <code>undo(model)</code>
<code>MainWindow</code> , <code>ModelCanvas</code> , <code>Tool</code> subclasses	GUI window, drawing, input tools	mutate model via <code>Command</code> ; draw via <code>element_graphics</code>
<code>element_graphics</code>	single source of truth for N/V/M diagram math	reads element end forces

5. Input and output structure

5.1 Input

Models can be created in the GUI (drawing tools + dialogs) or written as a plain-text input file parsed by `file_io.read_input_file`. The text format is section-based; key sections:

```
TITLE
<one-line title>

NODES n
<id> <x> <y>

MATERIALS n
<id> <E> [alpha] [density] [name]          # paired with a SECTIONS block
                                           # (legacy: <id> <A> <I> <E> [alpha]
[depth])
SECTIONS n
<id> <material_id> <A> <I> [depth] [name]

ELEMENTS n
<id> <node_i> <node_j> <section_id> [FRAME|TRUSS] [START|END|BOTH]

SUPPORTS n
<node_id> <ux> <uy> <rz> [settle_ux settle_uy settle_rz]

LOADS n
<node_id> <Fx> <Fy> <Mz>

MEMBER_UDL n / MEMBER_POINT_LOADS n
FRAME_TEMPERATURE n / TRUSS_TEMPERATURE n
```

Lines beginning with `#` are comments; bad tokens (unknown keys, malformed numbers, dangling references) raise descriptive `ValueError`s rather than being silently ignored. The GUI's native project format is `.spa.json`, which stores the model text plus the view/grid/selection state.

5.2 Output

The CLI prints a structured report with these stages: input echo, equation numbering (the `E` matrix and per-element `G` vectors), the assembled `K` and `F`, the solve (residual + nodal displacements), member end forces, support reactions, an equilibrium check, and a storage report. The GUI shows the same information graphically (see §7) and can export per-member internal forces to CSV. Reference outputs for the demo models are committed under `examples/final_demo/outputs/`.

6. Analysis procedure

The single-case pipeline (`main.run_analysis`) runs these steps:

1. **Read / receive the model** — from a file (`read_input_file`) or from the live GUI model.
2. **Number the DOFs** — `DofManager.from_model` assigns equation numbers and omits rotational DOFs where no member supplies bending stiffness.
3. **Validate** — connectivity (DFS for floating components), non-positive properties, zero-length members, coincident nodes.

4. **Assemble** — scatter each element's `k_global`, `p_global` into `K`, `F`; add nodal loads and (optionally) self-weight.
5. **Apply boundary conditions / settlements** — build the prescribed- displacement vector `D_s`.
6. **Solve** — $K_{ff} \cdot D_f = F_f - K_{fs} \cdot D_s$, after an SVD rank check.
7. **Recover member forces** — $q = k \cdot d - p$ in local coordinates per element.
8. **Compute reactions** — $R = K \cdot D - F$ at restrained DOFs.
9. **Check equilibrium** — sum element end forces at every free node; residuals are reported (typically $\sim 1e-13$).

For multiple load cases, `run_multi_case_analysis` runs step 1–9 per case and `MultiCaseAnalysisResult` superposes them for load combinations (exact, because the solver is linear).

7. Result visualization and post-processing

In the GUI, results are drawn as overlays on the model canvas, all toggled from the results panel:

- **Deformed shape** (with an adjustable visual amplification scale).
- **Internal-force diagrams** — bending moment (M), shear (V), and axial (N), selectable per result.
- **Support reactions** — forces and moments at restrained nodes.
- **Hover read-out** — the internal force value at the cursor's position along a member, and a per-member detail dialog with end forces and the full N/V/M plot.

Crucially, **all** diagram values flow through one routine,

`element_graphics.evaluate_internal_force / sample_internal_force /`

`internal_force_at`, so the hover read-out, the canvas diagram, and the detail dialog can never disagree. The $dm/dx = V$ sign convention is defined and unit-tested in exactly that one place.

Internal forces can also be exported to CSV at 21 stations per member for comparison against external tools such as SAP2000.

8. Verification examples

The solver was verified against closed-form hand calculations. Full details and the reproduction commands are in [docs/verification/final_verification.md](#); the headline result for the steel cantilever ($L = 4 \text{ m}$, tip load $P = 10 \text{ kN}$, $EI = 4074 \text{ kN}\cdot\text{m}^2$) is:

Quantity	Our Program	Reference (hand)	Difference	Status
Tip deflection $PL^3/3EI$ (m)	-5.236459e-02	-5.236460e-02	< 0.001 %	
Tip rotation $PL^2/2EI$ (rad)	-1.963672e-02	-1.963672e-02	< 0.001 %	
Fixed-end moment PL (kN·m)	40.0000	40.0000	0.0 %	
Base shear P (kN)	10.0000	10.0000	0.0 %	
Base reaction R_y (kN)	10.0000	10.0000	0.0 %	

A simply-supported beam (central point load) gives reactions 5 kN each, end rotation $PL^2/16EI = 0.0153413 \text{ rad}$, and midspan moment $PL/4 = 25 \text{ kN}\cdot\text{m}$, all matching the program. Both models pass the internal equilibrium check at machine precision. The verification exercises every stage of

the pipeline: stiffness assembly, boundary conditions, load-vector formation, solution, force recovery, and post-processing sign conventions.

The program is additionally backed by an automated test suite of ~670 tests (`pytest`) covering the engine, the command/undo layer, validation, and GUI smoke tests.

9. Limitations and possible improvements

Current limitations (stated honestly):

- **2D only** — no out-of-plane or 3D analysis; `nu` , `J` , and the section width/shape fields are stored but not consumed by the 2D solver.
- **Linear elastic only** — no material or geometric nonlinearity, no P-Δ, no buckling.
- **Static + modal** — no dynamic time-history or response-spectrum analysis.
- **No design checks** — the program reports demands, not code utilization.
- **Loads over rigid end-offset zones are not transferred to the joints.** When a frame member has rigid end offsets (`offset_i` / `offset_j`), member loads (UDLs, point loads) and self-weight are integrated over the **flexible span only**; the rigid zones are idealised as part of the joint and carry no distributed load. As a result, for a UDL applied across the full element the summed vertical reactions equal $w \cdot L_{flex}$, **not** $w \cdot L_{total}$. This is a deliberate, internally-consistent convention (point loads placed inside a rigid zone are rejected with an error rather than silently relocated), but it differs from tools such as SAP2000, which transfer the rigid-zone tributary load to the adjacent joint. Benchmark rigid-offset member-load totals with this in mind.
- **Two `validate_model` functions** (core vs. GUI UX) share a name across layers; correct by design but a potential point of confusion.
- **`element_graphics`** (a GUI-package module) holds the N/V/M mechanics rather than the engine; intentional and tested, but it places one piece of mechanics outside the engine package.

Possible improvements:

- Extend to 3D frames/trusses (the DOF and transformation machinery generalizes naturally).
 - Add P-Δ / geometric stiffness and basic nonlinear capability.
 - Add response-spectrum / time-history dynamics on top of the existing modal solver.
 - Sparse/banded storage for large models (the storage report already estimates the bandwidth savings).
 - Optional design-code post-processing (steel/concrete checks).
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10. References

1. K.-J. Bathe, *Finite Element Procedures*, Prentice Hall.
 2. R. C. Hibbeler, *Structural Analysis*, Pearson.
 3. A. Kassimali, *Matrix Analysis of Structures*, Cengage.
 4. W. McGuire, R. H. Gallagher, R. D. Ziemian, *Matrix Structural Analysis*, Wiley.
 5. NumPy and SciPy documentation — <https://numpy.org> , <https://scipy.org>
 6. Project source code and tests — `structural_analysis/` , `tests/` (this repository).
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Exporting to PDF. The repository already includes a Markdown → PDF renderer (docs/ render_proposal_pdf.py , using markdown + weasyprint). To produce report/ final_project_report.pdf , either:

1. **Reuse the existing renderer** (recommended). With the venv active and pip install markdown weasyprint:

```
bash python - <<'PY' import markdown
from weasyprint import HTML, CSS from pathlib import Path src =
Path("report/final_project_report.md") css = Path("docs/
render_proposal_pdf.py").read_text().split('CSS_TEXT = ""')
[1].split('""')[0] html =
markdown.markdown(src.read_text(encoding="utf-8"),
extensions=["tables", "fenced_code", "sane_lists"]) HTML(string=f"<!
doctype html><meta charset=utf-8><body>{html}</body>",
base_url="report").write_pdf("report/final_project_report.pdf",
stylesheets=[CSS(string=css)]) print("wrote report/
final_project_report.pdf") PY
```
2. **Or** use any Markdown editor (VS Code "Markdown PDF", Typora, or pandoc final_project_report.md -o final_project_report.pdf).

A weasyprint -rendered PDF is committed alongside this file when the tooling is available in the build environment; otherwise this Markdown source is the authoritative report.